HW7 · LLM clue-reasoning pipeline

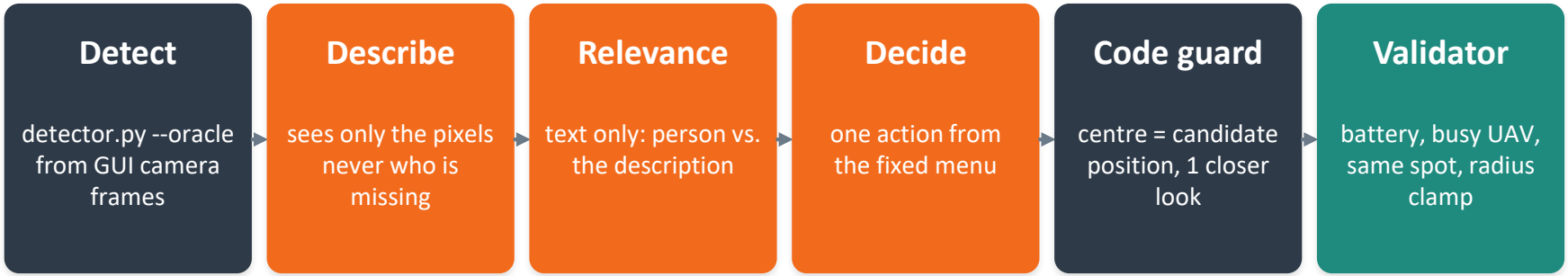
The model proposes. Code decides.

A search drone finds a cap, a backpack, a doll. Three Claude stages decide what each one means for this search. My HW6 validator decides whether anything flies.



Missing: Ethan, 8

Three stages, then the fence



mission/clues → my planner: inspect_closer → hover over the clue · converge_search → circle around the centre · ignore / log_only → nothing flies. **Every verdict goes on mission/action_result, marked decided_by: ai.**

My test set: Ethan, 8, near the north gate

Description, written before drawing
Navy cap with a grey front panel, green tee, denim shorts, white sneakers; small orange backpack with a green dinosaur patch.

Why each decoy
Beanie: right colour, wrong kind (near miss). Man's shoe: adult size. Doll: a child's toy, not his. Ball: tempting at a stadium.


RELEVANT · his cap


RELEVANT · his backpack


decoy · near miss: beanie


decoy · man's shoe


decoy · doll


decoy · soccer ball

Offline: every clue right, every run

15/15

Lily's set, 3 repeats
relevance and action

93%

action agreement
across repeats

6/6

my set (Ethan)
relevance and action

$0.02

per clue, three
Claude calls

The only disagreement: Lily's shoe went log_only, log_only, ignore (both acceptable). Closer looks: 8, all asked for by Describe. Code guard overrides: 0.

But these renders are sharp. The GUI camera in flight is not, and that is where it broke.

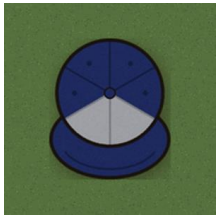
Traced failure: Relevance filed Ethan's cap away



offline render



in flight (GUI)



closer look 6×

Before

Describe ✓ honest: "blurry blue object, white-grey oval", confidence low, needs a closer look

Relevance ✗ first wrong stage: picked "blue denim shorts", said low, while its rationale named the cap

→ log_only: his cap ignored

Fix and after

Schema: possible_items before candidate_item, so it compares before it commits

Prompt: unclear but colour, size and outline fit → medium

→ inspect_closer → hover

→ zoomed: high → converge

In flight: every clue handled, the fence held

Ethan's cap	medium → inspect_closer	**approved: hovered 60 s**
cap, re-reported ×2	medium → inspect_closer	**rejected: busy, same spot**
Ethan's backpack	medium → inspect_closer	**approved: hovered 60 s**
beanie, ball, shoe, doll	low / none → log / ignore	**nothing flew**
2 converges at 26% battery	(inject_clue.py)	**rejected: reserve**

Never left to the model: coordinates, feasibility, a second look, a delivery.
One command runs it all: **bash hw07/run_demo.sh**